



ArcGIS Dashboards: Best Practices

David Nyenhuis
Soobin Choi

2026 ESRI USER CONFERENCE

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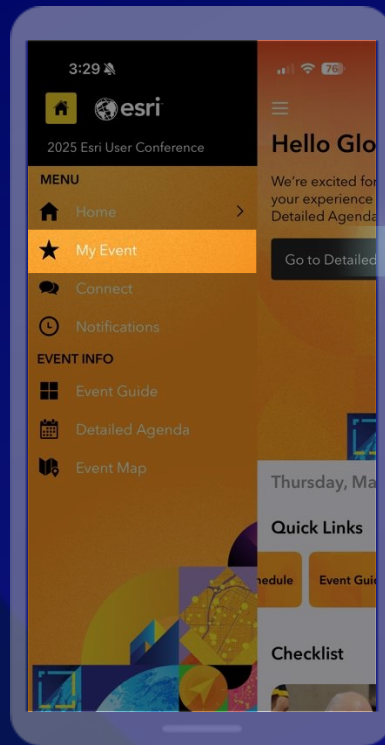
1. Connect your personal headphones/earbuds
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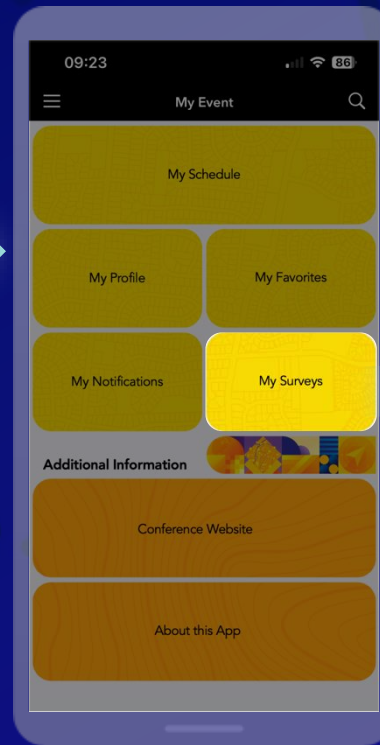
<https://link.esri.com/sessioncodes/mixhalotechpres>

Please share your feedback in the app

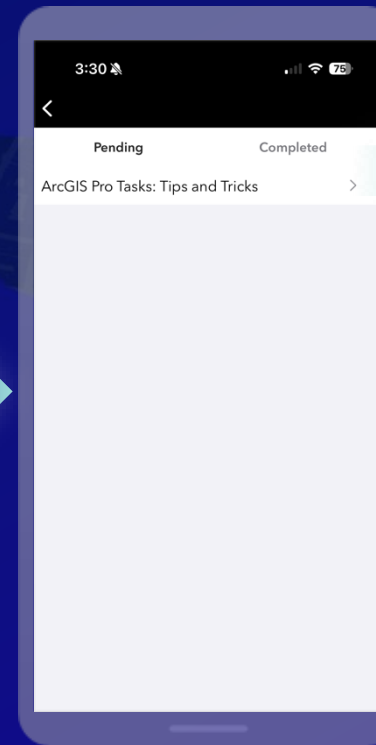
Select “My Event” in the event app main menu



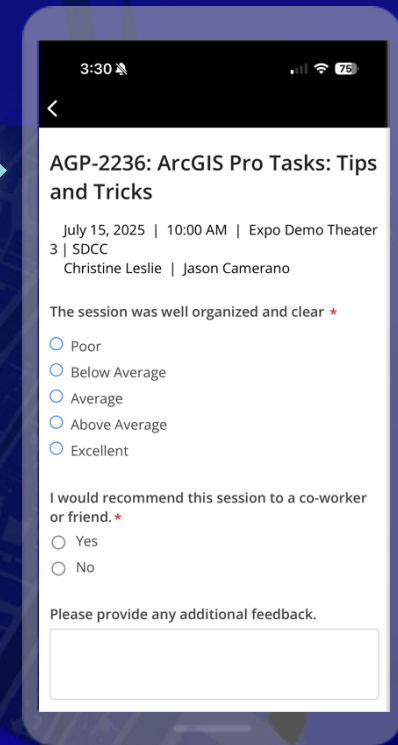
Select “My Surveys”



Select the desired survey



Complete the survey



Live first impression test

- 3 seconds to process a dashboard
- No analysis
- Just instinct

Collisions

632

4 Fatalities

39

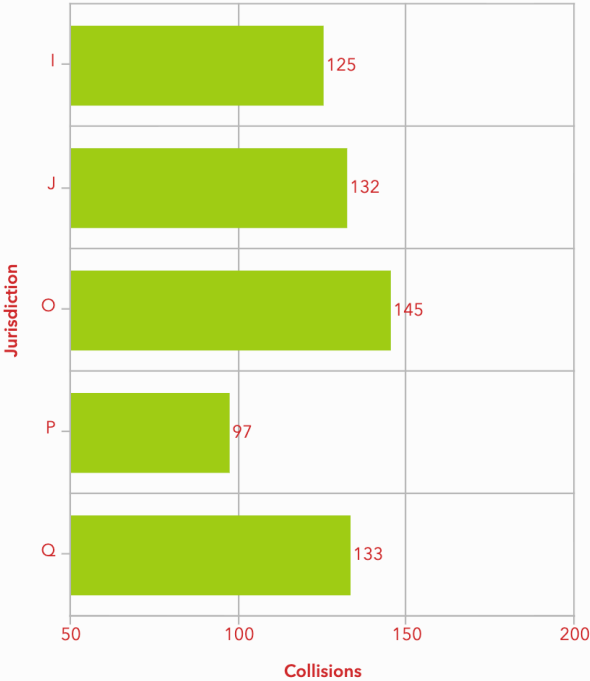
Pedestrians

Alcohol

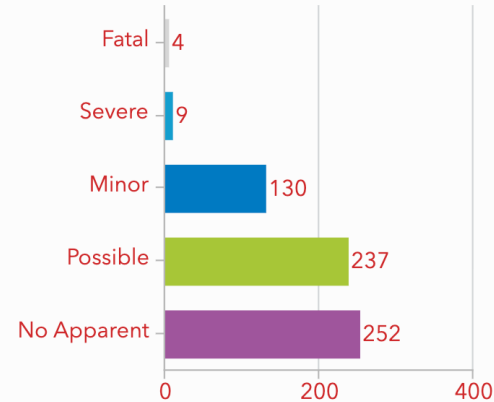
52

Involved

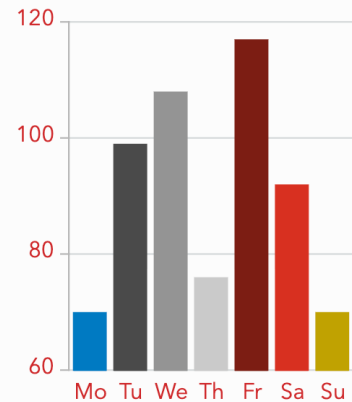
Collisions per Jurisdiction



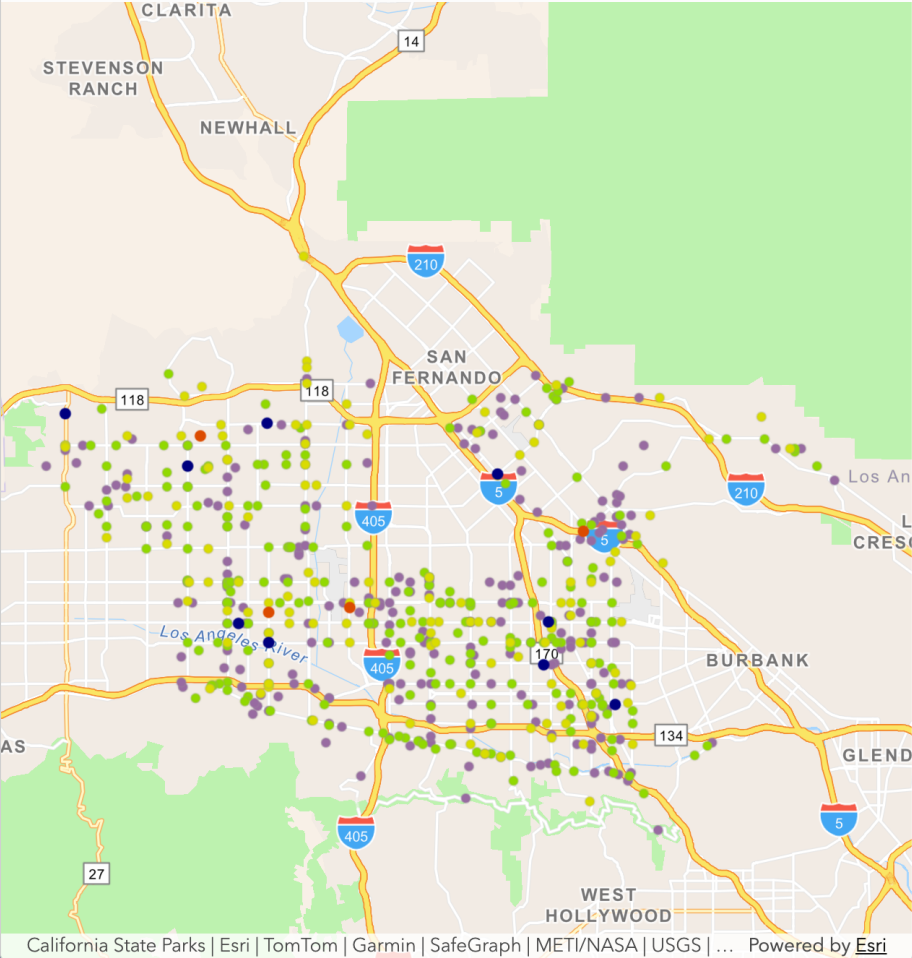
Collision Severity



Day of Week



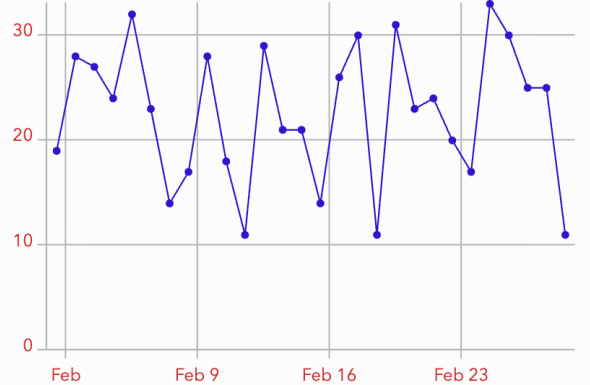
Collision Locations



High-Risk Intersections

Intersection	Collisions	Fatalities	Injuries
TAMPA AVE & SATICOY ST, LOS AN...	4	0	5
LAUREL CANYON BLVD & VANOWEN S...	4	0	3
SHERMAN WAY & COLDWATER CANYON...	4	0	1
ROSCOE BLVD & LOUISE AVE, LOS ...	3	0	6
RESEDA BLVD & SHERMAN WAY, LOS...	3	0	4
OSBORNE ST & GLENOAKS BLVD, LO...	3	0	2

Collisions per Day

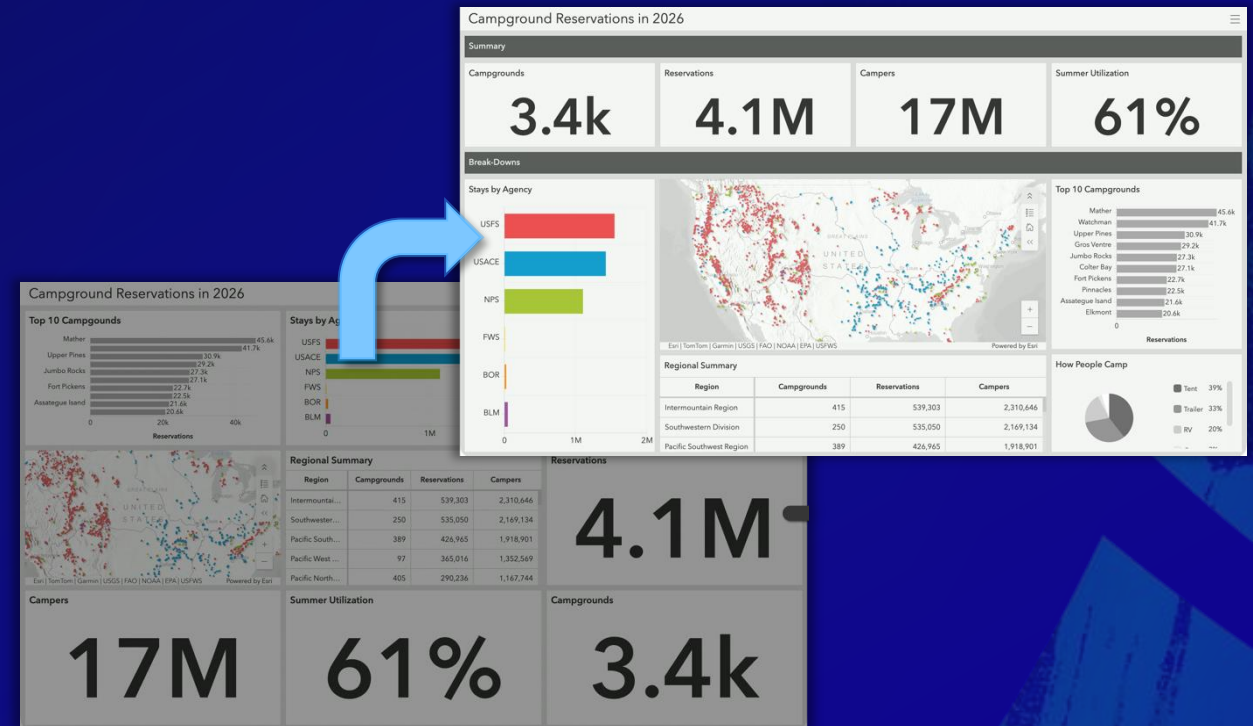


Reactions

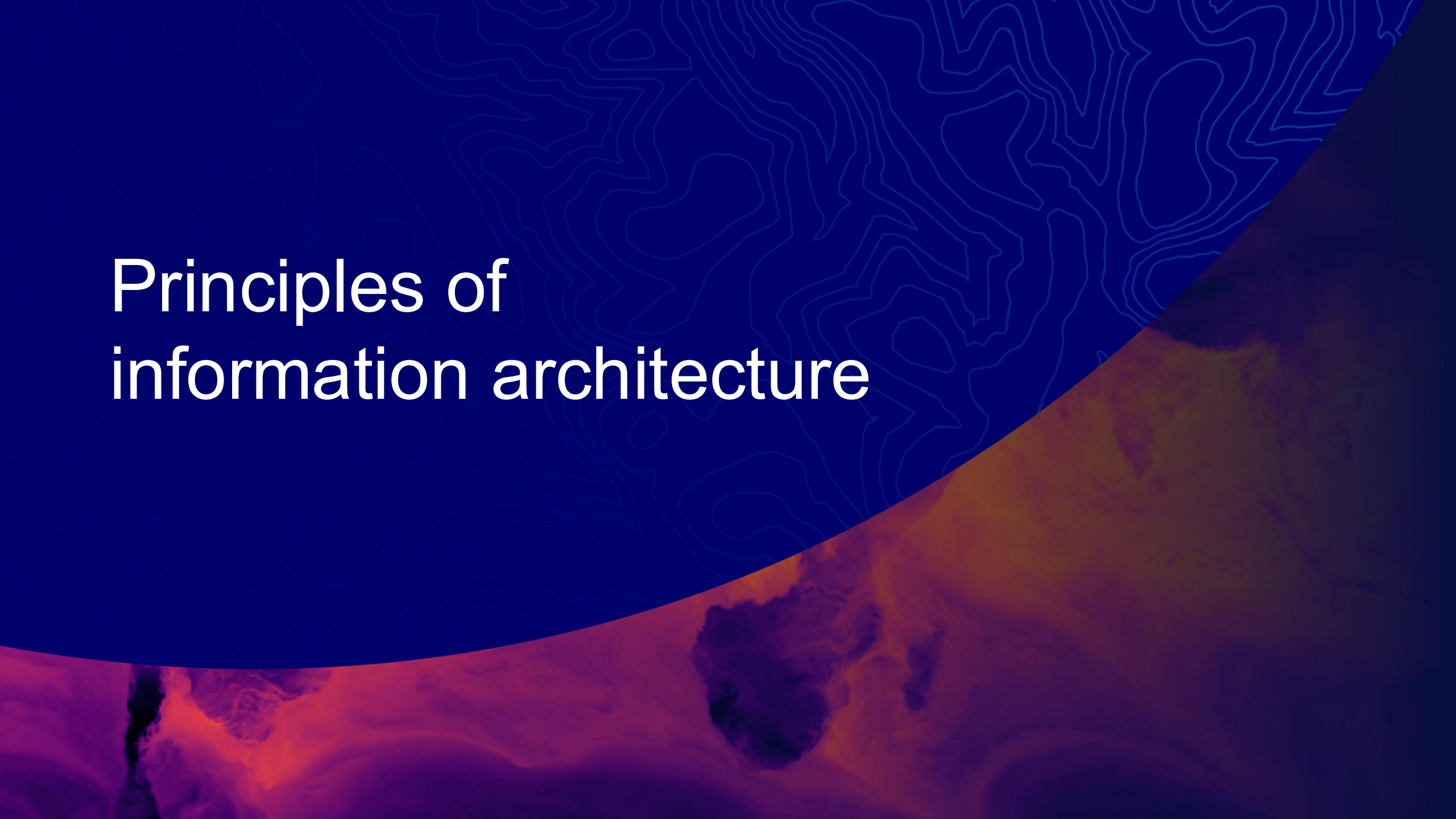
- What did you notice first?
- What did you think this dashboard is asking you to do?
- What felt important?
- Would you feel confident making a decision from it?

Agenda

- Principles of information architecture
 - Consistency
 - Clarity
 - Hierarchy
 - Context
- Roles dashboards can play
 - Inform users
 - Explore data
 - Explain insights
- Questions



Dashboard improvement exercises

The background of the slide is split into two main sections. The upper section is a solid dark blue color, overlaid with a white, intricate topographic map pattern of contour lines. A large, smooth, curved line separates this blue section from the lower section. The lower section features a vibrant sunset or sunrise scene with a gradient of colors from deep purple and blue at the bottom to bright orange and yellow near the horizon. Silhouettes of clouds and possibly trees are visible against the bright sky.

Principles of information architecture

Principles of information architecture

```
graph TD; C((Effective Understanding)) --- Consistency; C --- Clarity; C --- Hierarchy; C --- Context;
```

Consistency

Make it predictable

Clarity

Remove unnecessary effort

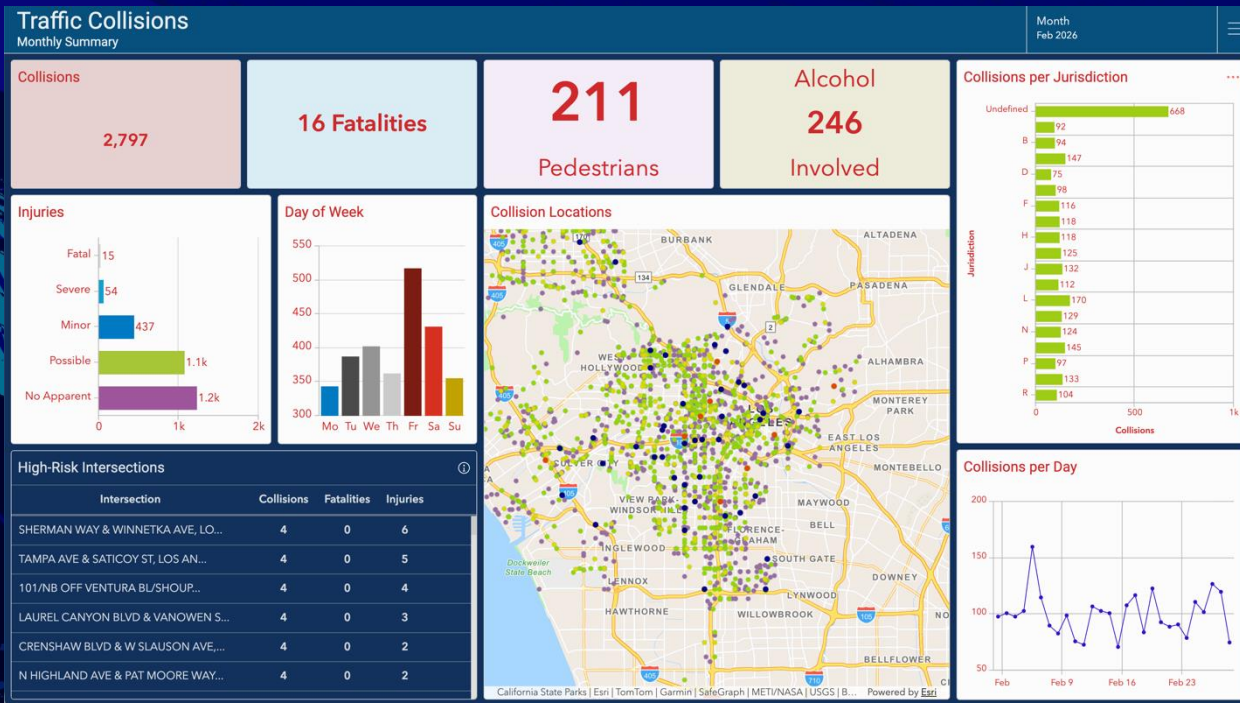
Effective
Understanding

Hierarchy

Guide attention to
what matters

Context

Make data meaningful

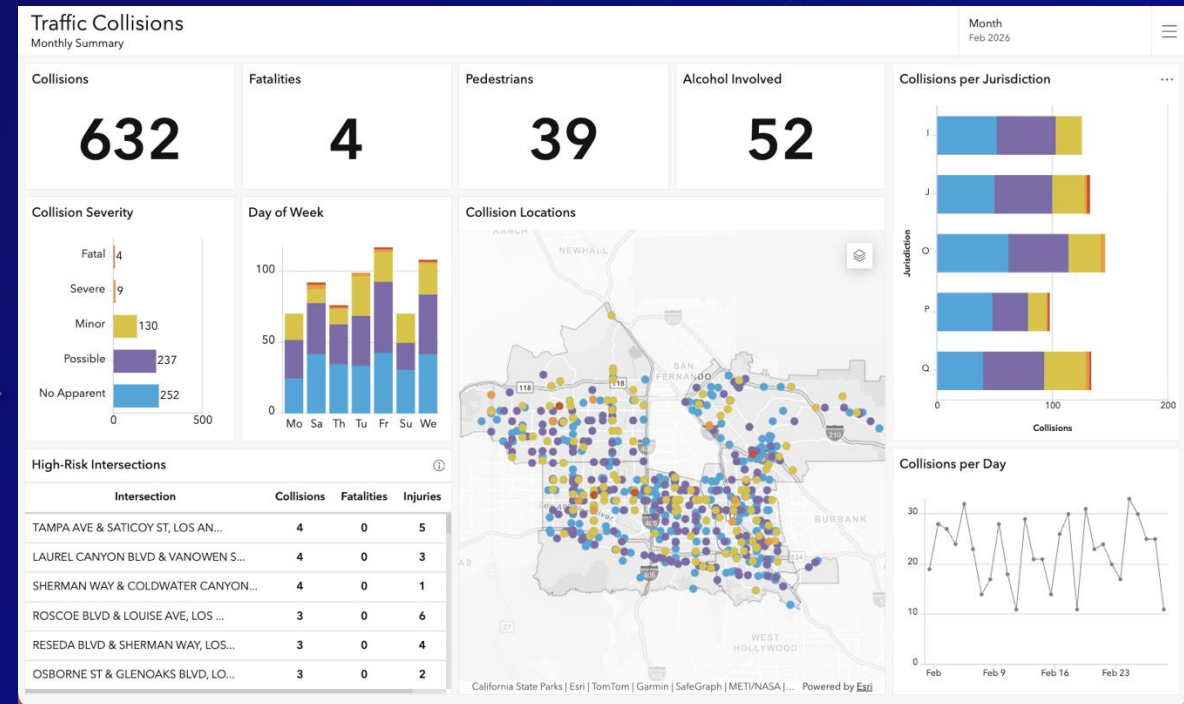
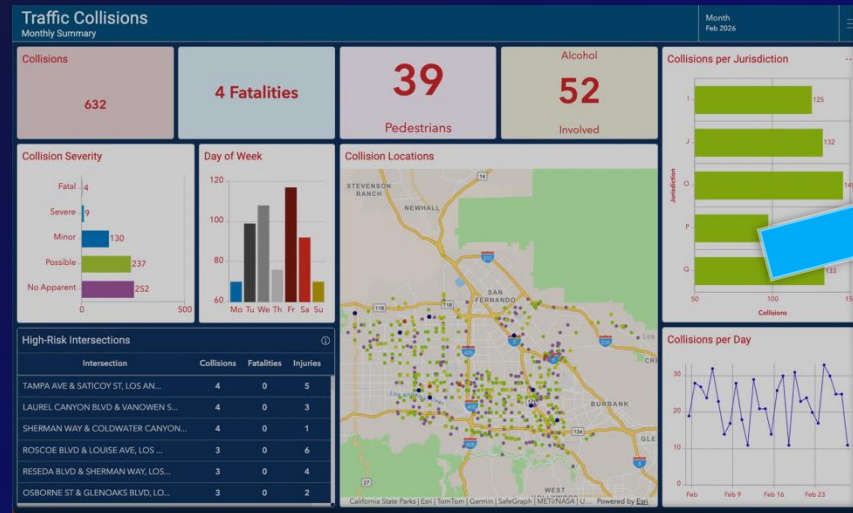


Consistency

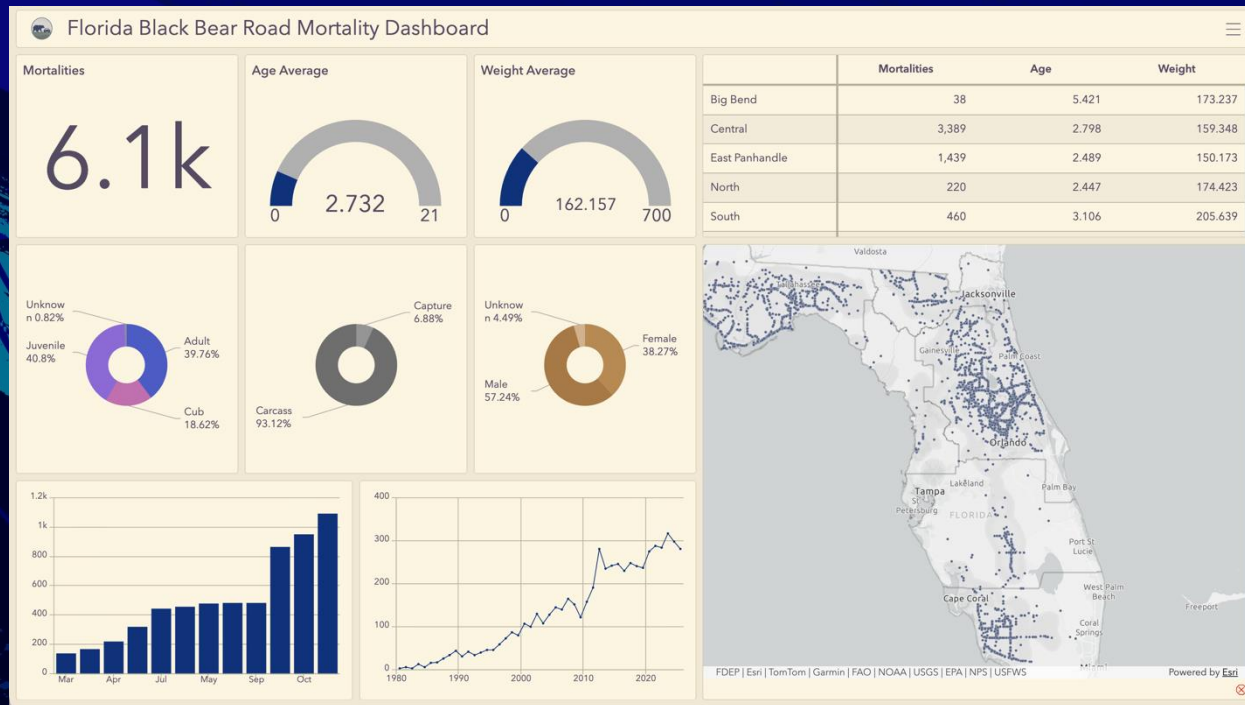
Dashboard Improvement 1

Consistency

Make it predictable



- Replace competing colors with a unified theme
- Align colors across maps, charts, and indicators
- Make insights easier to recognize at a glance

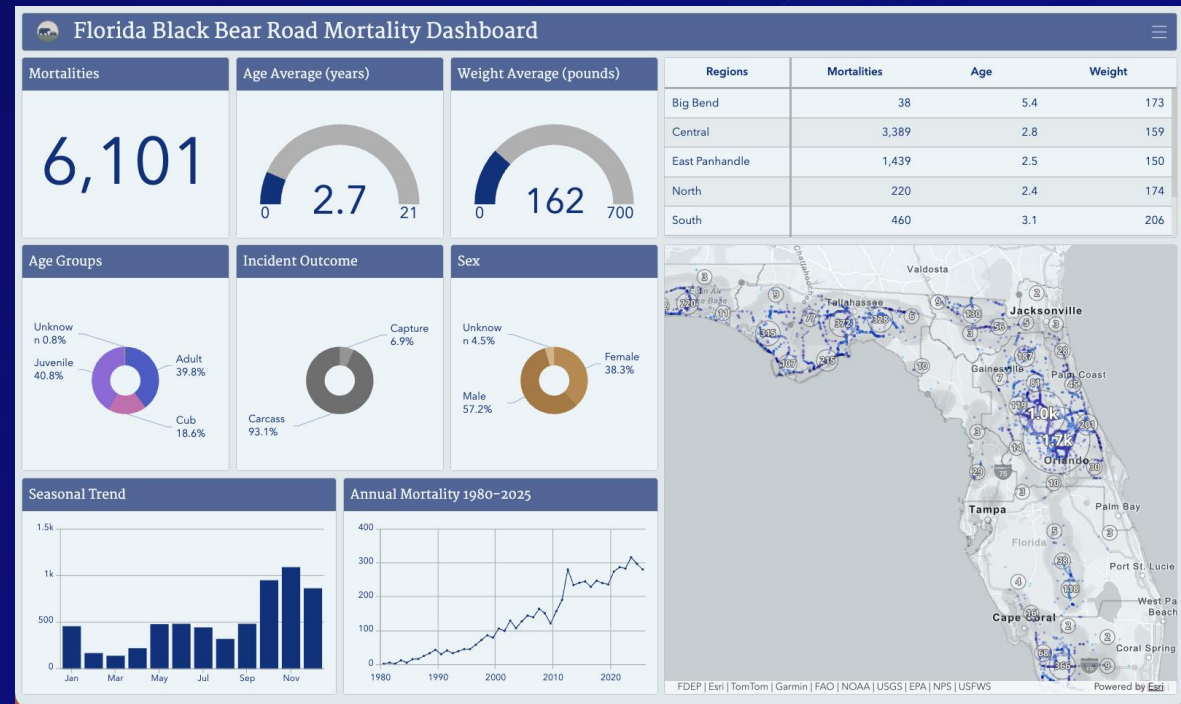


Clarity

Dashboard Improvement 2

Clarity

Remove unnecessary effort



- Replace guesswork with clear titles and readable values
- Sort information to match how people think and compare
- Surface spatial patterns so insights stand out naturally

Damage Assessments

Background Information

Welcome to the Hurricane Sampaúl Damage Assessment Dashboard. This dashboard provides a real-time operational view of assessment activities following Hurricane Sampaúl's landfall along Florida's Gulf Coast.

Use this dashboard to:

- Monitor damage assessment progress
- Track affected properties and infrastructure
- Identify priority response areas
- Review inspection status and findings
- Support recovery planning and resource allocation

All data displayed in this dashboard is fictional and intended for demonstration purposes only.
Last Updated: September 22, 2026, 8:15 AM EDT

Situation Summary

Hurricane Sampaul made landfall as a Category 4 storm, bringing widespread wind damage, coastal flooding, and power outages across multiple counties. Assessment teams continue to evaluate residential, commercial, and public assets to determine impacts and

Assessment Details

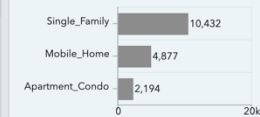
Report Number: **499-1964**
Report Date: **10/13/25, 6:05 AM**
Damage: **Major**
Address: **7874 Causeway Blvd S,
Saint Petersburg**
Dwelling Type: **Single_Family**
Comments:
*Visible waterline about 30 inches
high on side of house, single-story
water level same all round and
pressure wash stone in the front*

Report Number: 499-1973
Report Date: 10/13/25, 6:03 AM
Damage: Major
Address: 7119 Bay St, St Pete Beach
Dwelling Type: Single_Family
Comments:
Approx 6" water in living area.

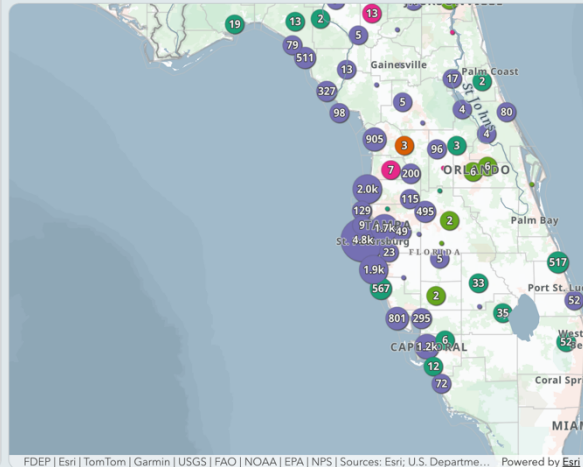
Report Number: 499-1914
Report Date: 10/13/25, 6:00 AM
Damage: Minor
Address: 11623 Caleb Ave, Port Richey
Dwelling Type: Mobile_Home
Comments:
MH damage waterline marked with blue tape

Report Number: 499-1946
Report Date: 10/13/25, 5:51 AM
Damage: NVD
Address: 7970 1st Ave S, Saint
Petersburg
Dwelling Type: Single Family

Dwelling Types



Collection Types



County
All counties

All counties

Non-visible damages

1,355

Affected

2,550

Minor damage

4,978

Major damage

8,268

Destroyed

394

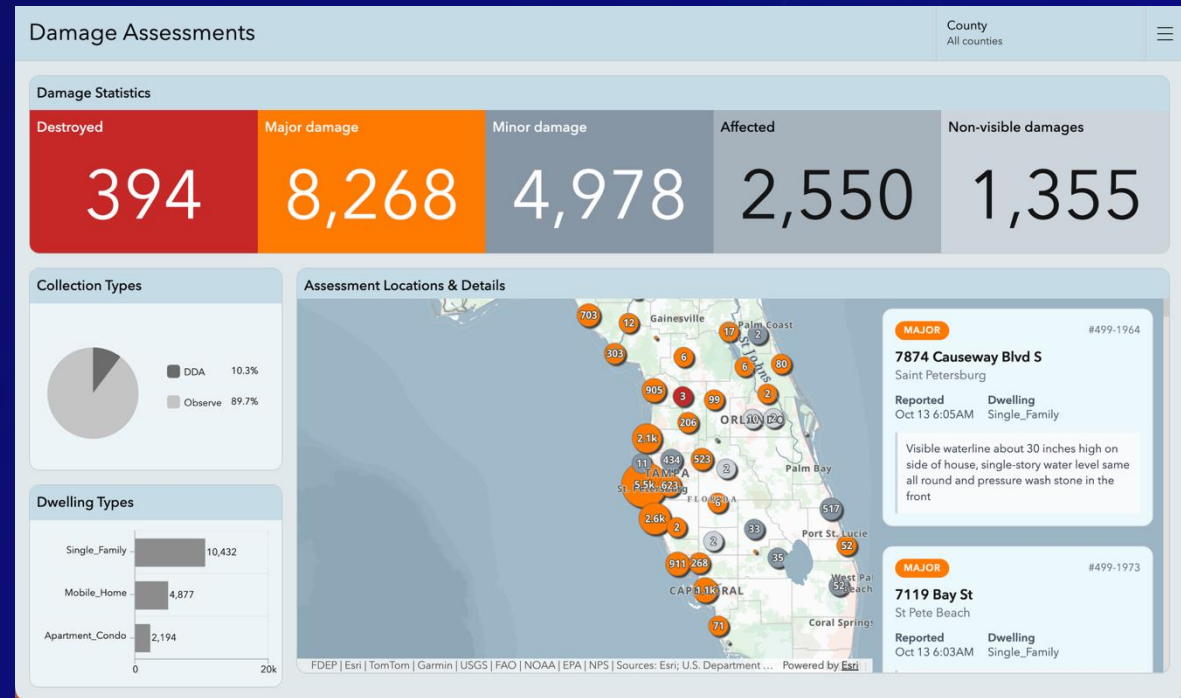
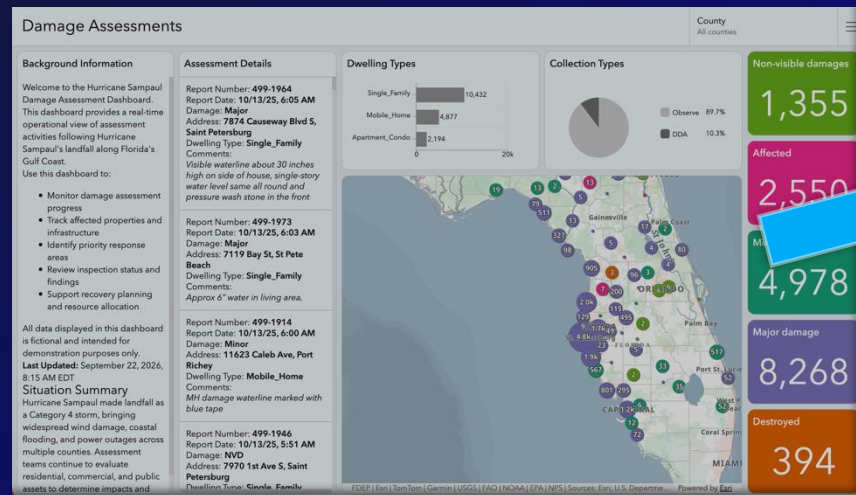
FDEP | Esri | TomTom | Garmin | USGS | FAO | NOAA | EPA | NPS | Sources: Esri; U.S. Department... Powered by Esri

Hierarchy

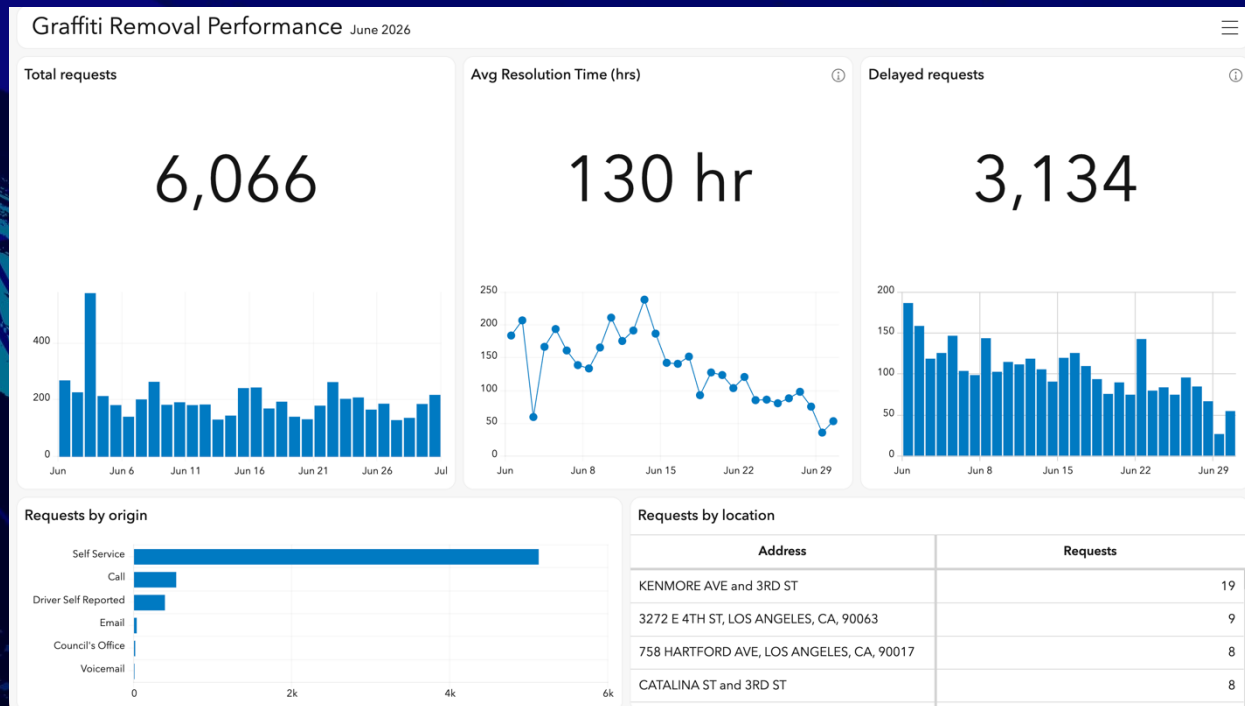
Dashboard Improvement 3

Hierarchy

Guide attention to what matters



- Lead users to key insights through purposeful layout
- Organize similar information into meaningful sections
- Emphasize what matters most with visual weight and color

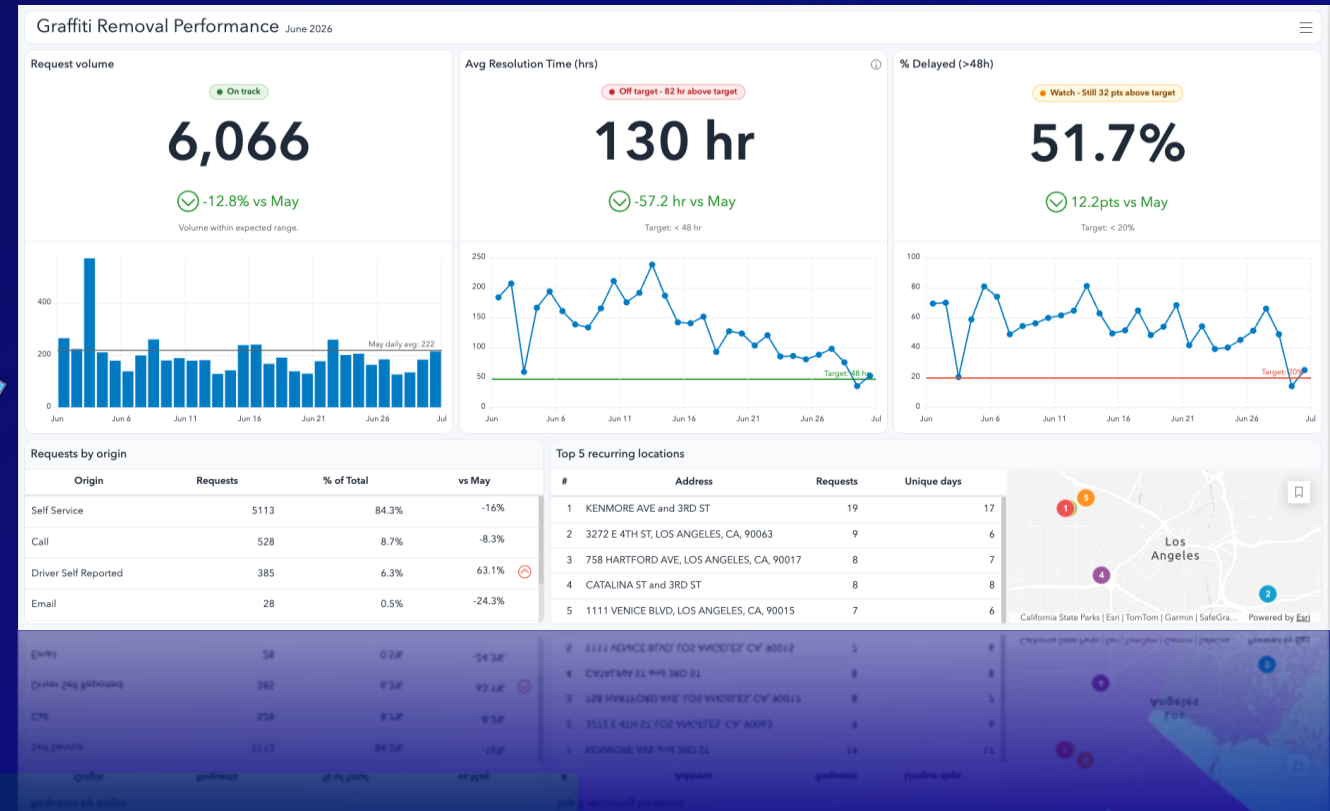


Context

Dashboard Improvement 4

Context

Make data meaningful



- Turns raw values into understanding
- Speeds up interpretation
- Enables informed decisions

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Roles dashboards
can play

Roles a dashboard can play

Inform

Users



- Immediate situational awareness
- Key metrics at-a-glance
- Minimal interaction required

Explore

Data



- Interactive data discovery
- Drill into patterns and outliers
- Answer your own questions

Explain

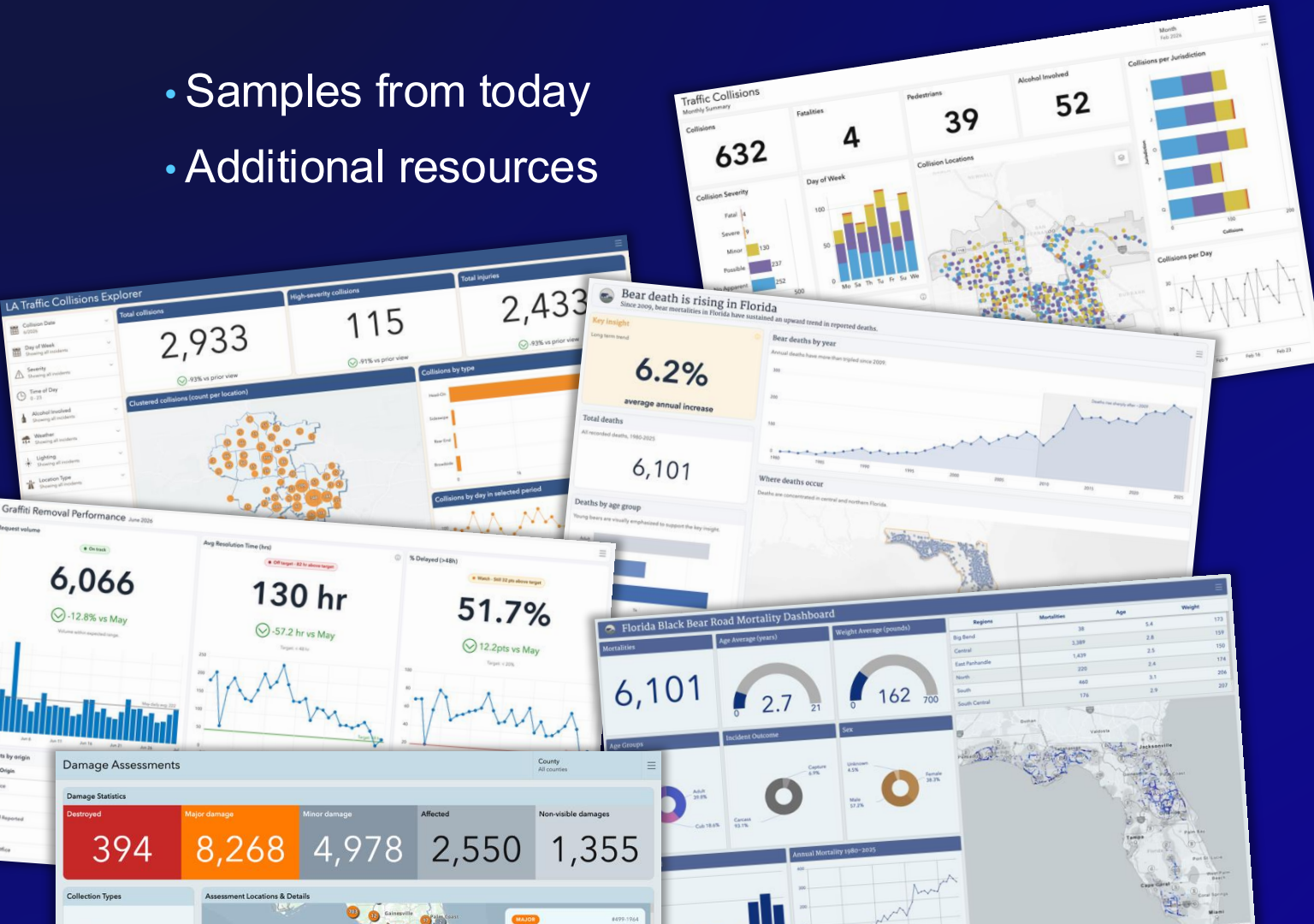
Insights



- Highlight what matters
- Reveal patterns and meaning
- Connect to decisions

Resources

- Samples from today
- Additional resources

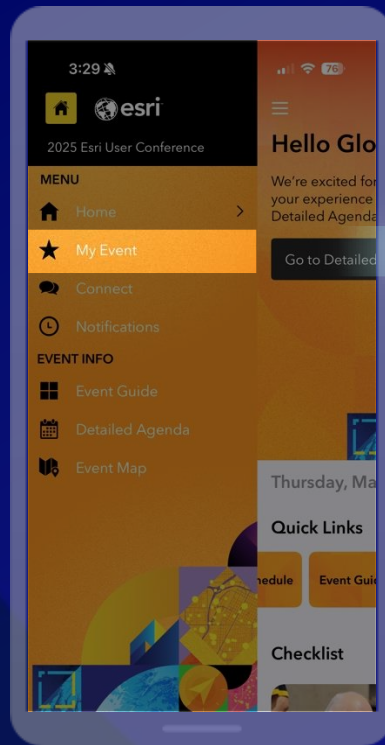


arcg.is/0fP5bz2

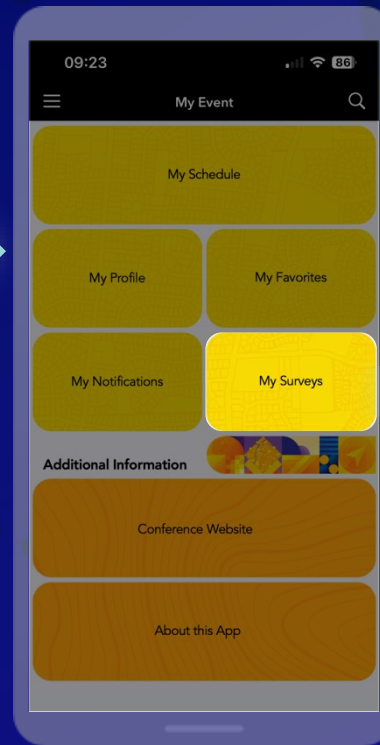


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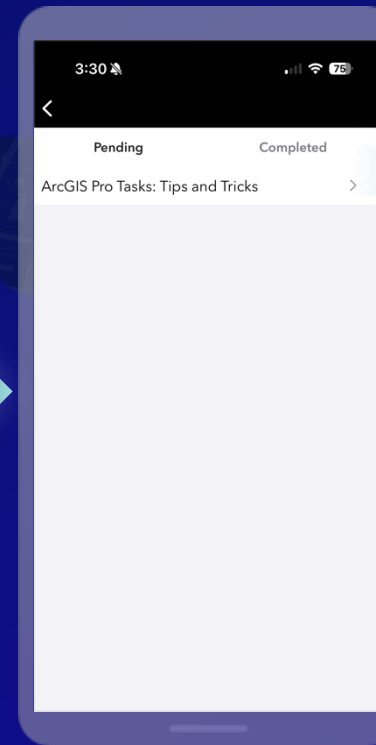
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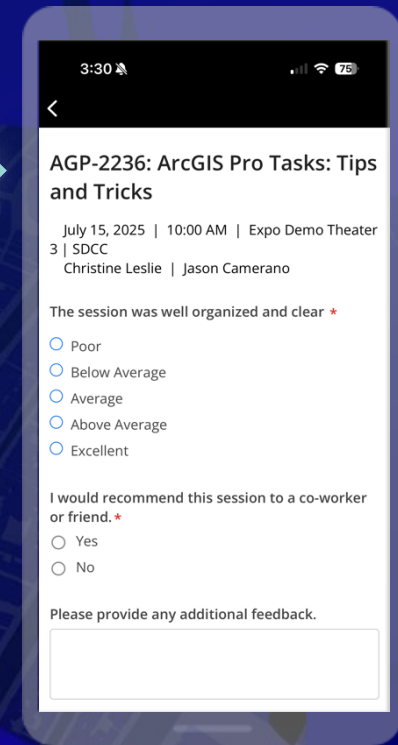
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Questions

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